0.0.1 Theoretic Motivation

Let the input $\boldsymbol{X}$ contain two underlying generative factors: a semantic factor reflecting the turbine operation $\mathfrak{s} \in \mathcal{S}$ (e.g., operating mode) and a detail factor $\mathfrak{g} \in \mathcal{G}$ (e.g., sensor-level variations or noise).

Let the encoder produce a normalized latent representation $\boldsymbol{z}$. The semantic mutual information is defined as

$$I(\boldsymbol{z}; \mathfrak{s}) = \mathbb{E}_{p(\boldsymbol{z},\mathfrak{s})} \left[\log \frac{p(\boldsymbol{z}, \mathfrak{s})}{p(\boldsymbol{z})p(\mathfrak{s})} \right], \tag{1}$$

measuring how much semantic content $\boldsymbol{z}$ retains. Similarly, the conditional mutual information

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}) \tag{2}$$

measures the amount of detail information contained in $\boldsymbol{z}$ after accounting for $\mathfrak{s}$. A desirable encoder should therefore maximize $I(\boldsymbol{z}; \mathfrak{s})$ and minimize $I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s})$ to alleviate the SL.

Theorem 1 (Semantic Enhancement and Detail Suppression). *Let $\boldsymbol{z} \in \mathbb{R}^d$ denote the normalized representation produced by the encoder, i.e., $\|\boldsymbol{z}\|_2 = 1$, and consider a contrastive learning framework with N anchor samples. Then the conditional mutual information between $\boldsymbol{z}$ and the detail factor $\mathfrak{g}$ satisfies*

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}) \leq C - \log N + \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}}}, \tag{3}$$

and

$$I(\boldsymbol{z}; \mathfrak{s}) \geq \log N - \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}}}, \tag{4}$$

where

$$\mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}}} = -\mathbb{E}_i \left[\log \frac{\exp(\mathrm{sim}(\mathfrak{s}_{i'}, \boldsymbol{z}_i))}{\sum_{j=1}^{N} \exp(\mathrm{sim}(\mathfrak{s}_j, \boldsymbol{z}_i))} \right], \quad i' = 1, \ldots, N \tag{5}$$

is an InfoNCE-type contrastive loss induced by a mode-matching mechanism under ideal wind turbine semantics,

$$C = \log \frac{2\pi^{d/2}}{\Gamma(d/2)}, \tag{6}$$

and $\Gamma(\cdot)$ denotes the Gamma function.

Proof Summary. The detailed proof is provided in Appendix I. $\square$

Furthermore, if the learned semantic representation s satisfies

$$\|s - \mathfrak{s}\|_2 \leq \varepsilon, \tag{7}$$

for a sufficiently small $\varepsilon > 0$, then there exists a constant $K > 0$ such that, by the proof of Appendix II,

$$\|\mathcal{L}_{\mathrm{NCE}_s} - \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}}}\|_2 \leq K\|s - \mathfrak{s}\|_2 \tag{8}$$

and

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}) \leq C - \log N + \mathcal{L}_{\mathrm{NCE}_s} \pm K\|s - \mathfrak{s}\|_2. \tag{9}$$

Thus, minimizing the loss $\mathcal{L}_{\mathrm{NCE}_s}$ associated with the true semantics $\mathfrak{s}$ can increase the mutual information $I(\boldsymbol{z}; \mathfrak{s})$ and tighten the upper bound of $I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s})$, thereby ensuring that the representation $\boldsymbol{z}$ increasingly reflects semantic structure.

A Proof of Theorem 1

According to (5) and the InfoNCE bound, the mutual information between $\boldsymbol{z}$ and the semantic factor $\mathfrak{s}_i$ satisfies

$$I(\boldsymbol{z}; \mathfrak{s}_i) \geq \log N - \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}_i}}. \tag{10}$$

By the chain rule of mutual information,

$$I(\boldsymbol{z}; (\mathfrak{s}_i, \mathfrak{g})) = I(\boldsymbol{z}; \boldsymbol{X}) = I(\boldsymbol{z}; \mathfrak{s}_i) + I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}_i). \tag{11}$$

Since $I(\boldsymbol{z}; \boldsymbol{X}) \leq H(\boldsymbol{z})$, we obtain

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}_i) \leq H(\boldsymbol{z}) - I(\boldsymbol{z}; \mathfrak{s}_i). \tag{12}$$

Substituting the InfoNCE bound yields

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}_i) \leq H(\boldsymbol{z}) - \log N + \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}_i}}. \tag{13}$$

Since $\|\boldsymbol{z}\|_2 = 1$, $\boldsymbol{z}$ lies on the unit hypersphere in $\mathbb{R}^d$, and its entropy satisfies

$$H(\boldsymbol{z}) \leq \log \frac{2\pi^{d/2}}{\Gamma(d/2)} = C. \tag{14}$$

Therefore,

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}_i) \leq C - \log N + \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}_i}}. \tag{15}$$

Equation (15) holds for all $i = 1, \ldots, N$. Since each $\mathfrak{s}_i$ represents semantic information within the wind turbine semantic space, increasing the mutual information $I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}_i)$ implies that more semantic information is captured in the representation $\boldsymbol{z}$. Hence, we have

$$I(\boldsymbol{z}; \mathfrak{g} \mid \mathfrak{s}) \leq C - \log N + \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}}}. \tag{16}$$

Similarly, the lower bound of $I(\boldsymbol{z}; \mathfrak{s})$ follows directly from the InfoNCE inequality. ■

B Proof of the Lipschitz Inequality

Let $\{s_i^+\}_{i=1}^N$ and $\{\mathfrak{s}_i^+\}_{i=1}^N$ be two sets of anchor vectors from different semantic Hilbert spaces, with corresponding positive and negative samples $\{\boldsymbol{x}_k\}_{k=0}^m$. Assume all vectors have norms bounded away from zero:

$$\|s_i^+\|, \|\mathfrak{s}_i^+\|, \|\boldsymbol{x}_k\| \geq c > 0, \quad \forall i, k, \tag{17}$$

and let $\tau > 0$ be the temperature parameter.

The NCE loss for anchor s_i^+ is defined as

$$\mathcal{L}_{\text{NCE}}(s_i^+) = -\frac{\text{sim}(s_i^+, \boldsymbol{x}_{k'})}{\tau} + \log\left(\sum_{k=0}^{m} \exp\left(\frac{\text{sim}(s_i^+, \boldsymbol{x}_k)}{\tau}\right)\right), \tag{18}$$

where $\text{sim}(s_i^+, \boldsymbol{x}) = \frac{s_i^+ \cdot \boldsymbol{x}}{\|s_i^+\|\|\boldsymbol{x}\|}$ is the cosine similarity, and $\boldsymbol{x}_{k'}$ is a positive sample of s_i^+.

Based on the Cauchy–Schwarz inequality and the Lipschitz property, for each k there exists a constant $C > 0$ such that

$$|\,\text{sim}(s_i^+, \boldsymbol{x}_k) - \text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_k)| \leq C\|s_i^+ - \mathfrak{s}_i^+\|_2, \quad \forall i, k. \tag{19}$$

The difference in NCE losses between s_i^+ and $\mathfrak{s}_i^+$ can be bounded as

$$\begin{aligned} \left|\mathcal{L}_{\text{NCE}}(s_i^+) - \mathcal{L}_{\text{NCE}}(\mathfrak{s}_i^+)\right| \leq\ & \frac{|\,\text{sim}(s_i^+, \boldsymbol{x}_0) - \text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_0)|}{\tau} \\ & + \left|\log \sum_{k=0}^{m} e^{\text{sim}(s_i^+, \boldsymbol{x}_k)/\tau} - \log \sum_{k=0}^{m} e^{\text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_k)/\tau}\right|. \end{aligned} \tag{20}$$

Using the Lipschitz property of the log-sum-exp function, the second term on the right-hand side can be further bounded by

$$\left|\log \sum_{k=0}^{m} e^{\text{sim}(s_i^+, \boldsymbol{x}_k)/\tau} - \log \sum_{k=0}^{m} e^{\text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_k)/\tau}\right| \leq \frac{\sum_{k=0}^{m} \left|e^{\text{sim}(s_i^+, \boldsymbol{x}_k)/\tau} - e^{\text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_k)/\tau}\right|}{\min(S_{s_i^+}, S_{\mathfrak{s}_i^+})}, \tag{21}$$

where

$$S_{s_i^+} = \sum_{k=0}^{m} e^{\text{sim}(s_i^+, \boldsymbol{x}_k)/\tau} \quad \text{and} \quad S_{\mathfrak{s}_i^+} = \sum_{k=0}^{m} e^{\text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_k)/\tau}. \tag{22}$$

Applying (19) and the Lagrange Mean Value Theorem to the exponential terms yields

$$\left|e^{\text{sim}(s_i^+, \boldsymbol{x}_k)/\tau} - e^{\text{sim}(\mathfrak{s}_i^+, \boldsymbol{x}_k)/\tau}\right| \leq \frac{C}{\tau} e^{1/\tau} \|s_i^+ - \mathfrak{s}_i^+\|_2. \tag{23}$$

Combining (20), (21), and (23), we obtain the Lipschitz bound for the i-th anchor:

$$\left|\mathcal{L}_{\text{NCE}}(s_i^+) - \mathcal{L}_{\text{NCE}}(\mathfrak{s}_i^+)\right| \leq K\|s_i^+ - \mathfrak{s}_i^+\|_2, \tag{24}$$

where

$$K = \frac{C}{\tau}\left(1 + e^{2/\tau}\right). \tag{25}$$

Finally, aggregating over all anchors $i = 1, \ldots, N$, we have

$$\left\|\mathcal{L}_{\mathrm{NCE}_s} - \mathcal{L}_{\mathrm{NCE}_{\mathfrak{s}}}\right\|_2 \leq K\left\|s^+ - \mathfrak{s}^+\right\|_2, \tag{26}$$

where $s^+ = [s_1^+, \ldots, s_N^+]$ and $\mathfrak{s}^+ = [\mathfrak{s}_1^+, \ldots, \mathfrak{s}_N^+]$. ■